



Password Store Protocol Audit Report

Version 1.0

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Table of Contents

- Table of Contents
- Protocol Summary
- Disclaimer
- Risk Classification
- Audit Details
 - Commit Hash:
 - Scope
 - Roles
- Executive Summary
 - Issues found
- Findings
 - High
 - * [H-1] Storing the password on-chain makes it visible to anyone and no longer private
 - * [H-2] PasswordStore::setPassword has no access controls, meaning a non-owner could change the password
 - Informational
 - * [I-1] The PasswordStore::getPassword natspec indicates a parameter that doesn't exist, causing the natspec to be incorrect.

Protocol Summary

A smart contract application for storing a password. Users should be able to store a password and then retrieve it later. Others should not be able to access the password.

Disclaimer

The Meharab team makes all effort to find as many vulnerabilities in the code in the given time period, but holds no responsibilities for the findings provided in this document. A security audit by the team is not an endorsement of the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the Solidity implementation of the contracts.

Risk Classification

		Impact		
		High	Medium	Low
Likelihood	High	H	H/M	M
	Medium	H/M	M	M/L
	Low	M	M/L	L

We use the CodeHawks severity matrix to determine severity. See the documentation for more details.

Audit Details

Commit Hash:

The commit hash that your findings correspond to: 2e8f81e263b3a9d18fab4fb5c46805ffc10a9990

Scope

```
./src/  
|-- PasswordStore.sol
```

Roles

- Owner: The user who can set the password and read the password.
- Outsides: No one else should be able to set or read the password.

Executive Summary

After spending 7 hours I have found 2 serious bugs and 1 informational bug.

Issues found

severity	number of issue found
high	2
medium	0
low	0
info	1
total	3

Findings

High

[H-1] Storing the password on-chain makes it visible to anyone and no longer private

Description: All data stored on chain is public and visible to anyone. The Password-Store::s_password variable is intended to be hidden and only accessible by the owner through the PasswordStore::getPassword function. I show one such method of reading any data off chain below.

Impact: Anyone is able to read the private password, severely breaking the functionality of the protocol.

Proof of Concept: The below test case shows how anyone could read the password directly from the blockchain. We use foundry's cast tool to read directly from the storage of the contract, without being the owner.

Impact: Anyone can set/change the stored password, severely breaking the contract's intended functionality **Impact:** Anyone can set/change the stored password, severely breaking the contract's intended functionality **Proof of Concept:** Add the following to the PasswordStore.t.sol test file:

Code

```
function test_anyone_can_set_password(address >randomAddress) public {
    vm.assume(randomAddress != owner);
    vm.startPrank(randomAddress);
    string memory expectedPassword = "myNewPassword";
    passwordStore.setPassword(expectedPassword);

    vm.startPrank(owner);
    string memory actualPassword = passwordStore.>getPassword();
    assertEq(actualPassword, expectedPassword);
}
```

Recommended Mitigation: Add an access control conditional to PasswordStore::setPassword.

```
if(msg.sender != s_owner){
    revert PasswordStore__NotOwner();
}
```

Informational

[I-1] The PasswordStore::getPassword natspec indicates a parameter that doesn't exist, causing the natspec to be incorrect.

Description:

```
/*
 * @notice This allows only the owner to retrieve the password.
 @> * @param newPassword The new password to set.
 */
function getPassword() external view returns (string memory) {}
```

The PasswordStore::getPassword function signature is getPassword() while the natspec says it should be getPassword(string).

Impact: The natspec is incorrect

Recommended Mitigation: Remove the incorrect natspec line.

```
/*
 * @notice This allows only the owner to retrieve the password.
-  * @param newPassword The new password to set.
+
 */
```